

M.TAYYAB SOHAIL

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Software Engineer | Computer Scientist | Full-Stack Developer

PROFESSIONAL SUMMARY

Computer Science graduate with 2 years of experience in software engineering and applied systems development. Experienced in designing and building full-stack applications, scalable backend services, and real-time web systems. Strong foundation in algorithms, data structures, and system design with practical exposure to cloud deployment and software architecture.

EDUCATION

Bachelor of Science in Computer Science

Sep 2021 – Jun 2025

FAST NUCES, Islamabad, Pakistan

- **Dean's List: Semesters 7 & 8** (top academic performer in final year AI & ML coursework)
 - **Key Subjects:** Data Structures & Algorithms, Algorithm Design & Analysis, Database Systems, Operating Systems, Computer Networks, Artificial Intelligence, Software Engineering, Information Security, Cloud Computing.
 - **Final Year Research Project: PhysioVision — Real-Time Human Pose Estimation for Adaptive Physiotherapy**
 - Investigated real-time human pose estimation for rehabilitation monitoring by designing a computer vision pipeline using MediaPipe and OpenCV to extract structured joint trajectories for movement analysis under real-world conditions.
 - Developed a retrieval-augmented language model framework that maps pose-derived features into context-aware physiotherapy feedback, enabling automated interpretation of patient movement patterns with approximately ~95% joint detection accuracy under real-time constraints.
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WORK EXPERIENCE

Software Engineer — Bitsmiths Studio

Jul 2025 – Present

- **AI SEO Automation Platform — (<https://seomaven.ai>)**
Developed an end-to-end Next.js-based SEO automation system integrating AI workflows for content generation and publishing pipeline automation, improving content generation throughput by 3× and reducing manual tracking overhead by 50%
- **Multi-Seller E-Commerce Marketplace — (<https://anina.app>)**
Designed a multi-tenant e-commerce system with modular backend architecture, implementing recommendation logic, payment workflows, and order management services, increasing user engagement by 40% and reducing support tickets by 70%.
- **Off-Campus Rental Marketplace — (<https://unibid.app>)**
Developed a real-time student rental marketplace using Next.js and Supabase, implementing WebSocket-powered bidding, secure role-based dashboards, and automated notification workflows, supporting **1,000+ concurrent users** with **<200ms** bid synchronization.
- **Real Estate Investment Platform — (<https://vestafi.co>)**
Engineered a full-stack fractional real estate investment platform with automated investment workflows, secure payment processing, and scalable marketplace architecture, reducing manual administrative effort while improving investment management efficiency.
- **Bitsmiths Studio Website — (<https://bitsmiths.studio>)**
Developed and optimized the company website using Next.js and TypeScript, implementing responsive UI, SEO optimization, and performance enhancements that improved page speed and organic search visibility while showcasing the company's products and services.

- **Human Resource Management System** — (<https://hrm.bitsmiths.studio>)
Built an enterprise HR management system with role-based access control, employee onboarding, leave management, payroll-ready data models, and secure document workflows, reducing HR administrative overhead through workflow automation.
Software Engineer — New Web Order — (<https://www.newweborder.co>) **May 2025 – Jul 2025**
- **Company Website:** Built and optimized a corporate website using Next.js, TypeScript, and Tailwind CSS, improving page performance by 40% through frontend optimization techniques and SEO restructuring
Software Engineer — Aawaz AI **Jun 2024 – Aug 2024**
- **BankIslami PEP Risk Intelligence System:** Built an NLP-powered news scraping and risk-ranking system to assess reputational and regulatory risk for Politically Exposed Persons (PEPs); enabled automated compliance monitoring for a banking client.
Software Engineer — IntegralX — (<https://snobots.vercel.app>) **Jan 2024 – May 2024**
- **White-Label AI Chatbot Platform:** Developed a MERN-based multi-tenant chatbot system using retrieval-augmented generation (RAG) architecture, enabling scalable deployment of AI assistants across multiple client instances and reducing onboarding time by 60%.

SKILLS & TOOLS

- **Languages & Frameworks:** React, Next.js, TypeScript, JavaScript, Tailwind CSS
- **Backend & Databases:** Node.js, Express.js, PostgreSQL, MongoDB, Supabase, Firebase, Pinecone
- **Cloud & Deployment:** AWS, Google Cloud, Docker, Kubernetes, Vercel, Coolify, Render
- **Automation & DevOps:** n8n, Trigger.dev, Git, GitHub, CI/CD
- **Payments & Third-Party APIs:** Stripe, PayPal, OpenRouter, DataForSEO, Google Maps API

CERTIFICATES

- AWS Academy Graduate — [Cloud Web Application Builder](#) (Dec 2024)
- AWS Academy Graduate — [Cloud Data Pipeline Builder](#) (Dec 2024)
- AWS Academy Graduate — [Cloud Foundations](#) (Dec 2024)

EXTRACURRICULAR AND LEADERSHIP

- **Technical Vice Head, Google Developer Student Club (GDSC):** Led a 10-member Platform Development team; guided development of a vocal aid system for hearing-impaired individuals.
- **Operations Specialist, TEDx FAST NUCES:** Organized one of the university's largest annual events with approximately 5,000 attendees.